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FEDERAL SPECIFICATION
ENAMEL
(IN PRESSURIZED DISPENSERS)

The General Services Administration has authorized the use of this federal specification for all Federal Agencies.

1. **SCOPE.** This specification covers an enamel for general-purpose interior and exterior application.

2. **APPLICABLE DOCUMENTS**

2.1 **Government Publications.** The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards:

FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials: Methods of Inspection, Sampling, and Testing.
FED-STD-313 - Material Safety Data Sheets, Preparation and Submission of

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 1210 - Fineness of Dispersion of Pigment-Vehicle Systems.
D 1296 - Odor of Volatile Solvents and Diluents.
D 1308 - Effect of Household Chemicals on Clear and Pigmented Organic Finishes.
D 2369 - Volatile Content of Coatings.
D 2697 - Volume Nonvolatile Matter in Clear or Pigmented Coatings
D 2805 - Hiding Power of Paints by Reflectometry.
D 3335 - Determination of Low Concentration of Lead in Paint by Atomic Absorption Spectroscopy.
D 3624 - Low Concentrations of Mercury in Paint by Atomic Absorption Spectroscopy
E 260 - General Gas Chromatography Procedures

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. **REQUIREMENTS**

3.1 **Materials.** The enamel shall consist of any suitable combination of resins, pigments, solvents, and propellants to meet all requirements of this specification.

3.1.1 **Prohibited materials.** When tested as specified in Table II and 4.3.1, the enamel shall not contain benzene, chlorinated solvents or ethylene-based glycol ethers and their acetates. Mercury and hexavalent chromium compounds shall not be detected. Lead content shall not exceed 0.06% by weight of the nonvolatile content. ACGIH carcinogens and ACGIH suspected carcinogens shall not be used. Halogenated solvents and propellants shall not be present when tested as specified in Table II.

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3.2 **Dispenser.**

3.2.1 **Container.** Containers shall be commercial pressure types generally known as aerosol containers of such construction as to assure acceptance of the finished package by common carriers operating in interstate commerce. The capacity shall be a 475 mL (16-ounce) container with outage in accordance with generally recognized safe commercial practice.

3.2.2 **Dispensing valve.** The valve shall have a spray head which can be removed without releasing pressure from the aerosol. The valve shall have a metering orifice accessible to cleaning with a common pin after removal of the spray head. The removable spray heads shall contain orifices of such dimensions as to produce spraying properties as required in 3.4.4.

3.2.3 **Agitator.** Each dispenser of the pigmented material shall contain one or more agitators capable of dispersing the pigment by normal shaking. The agitators shall be made of materials which shall not react with or contaminate the material or container.

3.2.4 **Cover cap.** The valve shall be protected from accidental functioning and damage by a press fitting metal or plastic cover cap which shall be removable and replaceable. The color of the cover cap shall be an approximate match to the color of the enamel in the container with the exception of colorless lacquer. The color of the cap containing the colorless enamel shall be clear.

3.2.5 **Interior surface.** The contents shall not react with nor be reacted upon by the interior surface of the dispenser or any of the dispenser components.

3.3 **Quantitative requirements.** The aerosols shall meet the quantitative requirements specified in Table I.

TABLE I. Quantitative requirements

Characteristics	Min.	Max.
Fineness of grind (except gold, silver and flat black)	7	--
60° specular gloss (for gloss colors)	80	--
60° specular gloss (for semigloss colors)	40	60
60° specular gloss (for flats)	--	10
Total solids by volume, mL		
aluminum and gold	45	--
all others	70	--
Contrast ratio at .025 mm(1 mil)		
FED-STD-595 color numbers		
12199, 13655, 23655	0.80	
11105, 11136, 11140, 12197, 13538, 21105, 21136, 23538	0.85	--
Other Colors		
Reflectivity 80 and higher	0.94	--
76-79	0.95	--
72-75	0.96	--
68-71	0.97	--
61-67	0.98	--
60 and below	0.99	--
Drying time		
tack free	---	15 min.
dry through	---	2 hours

3.4 **Qualitative requirements.**

3.4.1 **Storage stability.** At all times during the year following the date of acceptance, the aerosols must be usable. Sampling shall be in accordance with method 1022 of FED-STD-141 except that a minimum of two dispensers shall be taken from every 1000 dispensers. Inspection level and AQL level shall be in accordance with 4.2.1. Usability is determined by:

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(a)Spraying properties (see 3.4.4).
(b)Condition in container (see 3.4.3).
(c)Color match (see 3.4.2.).
(d)Drying time (see Table I).
(e)Valve operation (see 3.4.7).

3.4.2 **Color.** When tested as specified in 4.3.8, the color match the color specified within a delta E of +/- 1.75.

3.4.3 **Condition in container.** The enamel shall be sprayable after vigorously shaking the aerosol can by hand for 1 minute. The spray shall be free from seeds, specks, and agglomerates and otherwise produce a smooth and uniform film when tested as specified in 4.3.7.

3.4.4 **Spraying properties.** When tested as specified in paragraph 4.3.7 the content of the aerosol container shall be sprayable when stabilized at temperatures from 60° to 90°F. A minimum of 98 percent by weight of content shall be sprayable in a pattern such as to deposit a uniform coating on a smooth vertical surface. The film shall dry free of orange peel, sags, specks, or any other surface irregularities.

3.4.5 **Odor.** When tested as in Table II, the odor shall be normal for the volatiles used in the enamel. After the drying of the films there shall be no residual odor.

3.4.6 **Immersion resistance.** When tested as specified in 4.3.8, immediately after immersion of the panel, the film shall show no blistering or other defects. There shall be no softening of the film after 24 hours drying.

3.4.7 **Valve operation - freedom from clogging.** The valve shall operate freely without excessive finger pressure. It shall close immediately upon release of finger pressure. The spray shall be continuous without sputtering or interruption. The valve shall not clog nor collect heavy deposits of coating around the orifice when tested as specified in 4.3.9

3.4.8 **Flexibility (cold cracking).** The materials shall show no evidence of cracking when subjected to the flexibility test as specified in 4.3.10.

3.4.9 **Self-lifting.** When tested as specified in 4.3.11, the enamel shall show no self-lifting.

3.4.10 **Accelerated yellowing.** Colors with a reflectance over 80 shall have a maximum yellowness index difference of 0.07 when tested as in Table II.

3.5 **Material Safety Data Sheet.** A Material Safety Data Sheet (MSDS) shall be submitted in accordance with FED-STD-313 (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein using facilities approved by the Government. The Government reserves the right to perform any of the inspections set forth herein when deemed necessary to assure that the enamel conforms to prescribed requirements.

4.2 **Classification of inspections.** Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.2.1).

4.2.1 **Inspection of preparation for delivery.** A random sample of filled containers shall be selected in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 2.5 defects per hundred units.

4.3 **Quality conformance inspection.** The enamel shall be tested in accordance with the methods specified in Table II and as otherwise specified herein to determine compliance with the requirements of section 3 and Table I. Unless otherwise specified, all tests shall be conducted at conditions specified in section 9 of FED-STD-141. Failure of any test requirement shall be cause for rejection of the lot from which the sample was taken.

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Table II Index of Tests

Test	Requirement Paragraph	FED-STD-141 Method	ASTM Test Method	Test Paragraph
Prohibited Materials				
Solvent and Propellant Analysis	3.1.1	-----	E260	-----
Lead Content	3.1.1	-----	D 3335	4.3.1.1
Hexavalent Chromium Content	3.1.1	-----	-----	4.3.1.2
Mercury content	3.1.1	-----	D 3624	-----
Fineness of Grind	Table I	4411	-----	-----
60° Specular Gloss	Table I	6101	-----	4.3.2
Nonvolatile Content	Table I	----	D 2369	4.3.3
Contrast Ratio	Table I	4122	D 2805	4.3.4
Drying Time	Table I	4061	-----	4.3.5
Color	3.4.2	4250	-----	4.3.6
Spraying Properties	3.4.4	-----	-----	4.3.7
Odor	3.4.5	-----	D 1296	-----
Immersion Resistance	3.4.6	2011	D 1308	4.3.8
Valve operation	3.4.7	-----	-----	4.3.9
Flexibility	3.4.8	2012, 6221	-----	4.3.10
Self-lifting	3.4.9	-----	-----	4.3.11
Accelerated yellowing	3.4.10	6132	-----	-----

4.3.1 **Prohibited materials**

4.3.1.1 **Lead content.** Determine lead in accordance with ASTM D 3335 or by the use of an x-ray fluorescence spectrometer capable of determining lead at a minimum range of 0.03 through 1.0 percent mass of nonvolatile with an accuracy within plus or minus five percent. The x-ray method shall be used in case of dispute. Determine compliance with 3.1.1.

4.3.1.2 **Hexavalent chromium content.** Add 5 mL 25 percent aqueous KOH to 1/2 gram extracted pigment in a centrifuge tube. Agitate by shaking and centrifuge. A yellow color in the supernatant liquid indicates the presence of hexavalent chromium. Observe for compliance with 3.1.1.

4.3.2 **Specular gloss.** Spray the enamel onto a glass panel until complete hiding is obtained and determine gloss in accordance with method 6101 of FED-STD-141. Examine for compliance with Table I.

4.3.3 **Nonvolatile content.** Weigh a full container and carefully puncture the top of the can (the container may be cooled to lower pressure) and allow the gas to escape. Cut the top completely open and allow the can and contents to equilibrate at room temperature. Stir the contents to allow all propellants to escape. Transfer the entire contents to a tared container (capable of being sealed), washing the can, agitator and top thoroughly with a suitable solvent. Dry the can, agitator and top and reweigh to determine the net contents. Determine total solids percentage of the container contents in accordance with ASTM D 2369. Run tests in duplicate. Determine compliance with Table I. (This test can be conducted using the concentrate prior to packaging unless there is a dispute.)

4.3.4 **Contrast ratio.** Using the container contents from test 4.3.3, draw down a film of the enamel with a film applicator on a black and white Leneta card to a dry film thickness of 0.026 mm (1 mil). Determine the contrast ratio in accordance with ASTM D 2805. and examine for compliance with Table I. (This test can be conducted using the concentrated prior to packaging unless there is a dispute.)

4.3.5 **Drying time.** Prepare a panel as described in 4.3.7, time by method 4061 of FED-STD-141. Determine compliance with Table I.

4.3.6 **Color.** The enamel coating shall be sprayed on a 8 by 15 cm (3 by 6 inch) metal panel until complete hiding is obtained, and compliance with 3.4.2 determined by method 4250 of FED-STD-141.

4.3.7 **Spraying properties.** Weigh the full container of enamel. Spray a 8 by 15 cm (3 by 6 inch) steel panel (prepared in accordance with method 2011 of FED-STD-141). Set the panel at an angle of approximately 40 cm (15 in.) from the vertical. Hold the dispenser in an upright parallel position to the panel

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at a distance of 25 to 30 cm (10 to 12 in.) and with the valve fully open, move the dispenser horizontally from one side of the panel to the other using a stroking motion. Spray in overlapping horizontal strokes from the top to the bottom of the panel. Apply two thin coats (each coat approximately .013 mm (0.5 mil) dried film thickness) one after the other. The second thin coat can be applied by spraying vertically in overlapping strokes from side to side. After application of the second thin coat, place the panel in a vertical position, to dry for 5 minutes, and examine the film for smoothness of flowout, freedom from specks, bubbles, craters or any apparent abnormalities. Determine compliance with 3.4.4. Continue spraying the content to completion on a flat surface and note the spraying properties at intervals. Weigh the container to determine the weight of enamel discharged. Open the container and rinse with solvent to remove any residual paint. Dry the can at 105°C (221°F) for one hour, cool, and weigh to determine the weight of the enamel in the full container. Determine the percentage of enamel discharged and examine for compliance with 3.4.4.

4.3.8 **Immersion resistance.** Prepare two panels as described in 4.3.7. On the panel to be immersed, all exposed uncoated metal surfaces shall be coated with wax or other suitable coating. Air dry for 48 hours, then immerse in room temperature distilled water for 16 hours in accordance with ASTM method D 1308. Immediately observe for compliance with 3.4.6.

4.3.9 **Valve operation.** Spray from the dispenser with the valve fully open for 15 seconds each time, once per day, on any 7 days over a period of 10 days. Observe for compliance with 3.4.7. (After each spraying period, the valve shall be cleared according to manufacturers directions.)

4.3.10 **Flexibility.** The panel shall be as specified in FED-STD-141 method 2012. Prepare the panel as specified in 4.3.7 and air dry for 48 hours. Bend the panel over a 1/4 inch diameter rod as specified in method 6221 of FED-STD-141, and examine for compliance with 3.4.8.

4.3.11 **Self-lifting.** Spray the enamel on two panels as specified in 4.3.7. Recoat one panel after 4 hours air drying and the other after 48 hours air drying. Evaluate for compliance with 3.4.9.

5. PREPARATION FOR DELIVERY (see 3.2)

5.1 **Packaging, packing and marking.** Packaging, packing and marking shall be as stated in contract solicitation or order. (see 6.2)

6. NOTES

6.1 **Intended use.** Enamels in pressurized dispensers provide a quick convenient method for painting small objects or areas with a general-purpose coating. They can be applied on metal, wood, paper, linoleum, plaster, masonry, stone, glass, leather, and fiber; over bare, or previously painted surfaces. They may be used on, furniture, machinery, tools, radiators, window frames, trim, shutters, garden implements, and electrical and electronic equipment.

6.2 **Ordering data.** Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Color required (see 3.4.2).
(c) Packaging, packing and marking required (see 5.1).
(d) Availability of color chips if not standard (see 3.4.2).
(e) Instructions and address for submission of MSDS (see 3.5).

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